

Order Parameters Analysis

1 Order Parameters

To characterize the orientational organization of elongated cells, the following global order parameters are recorded.

The global nematic order is defined as

$$Q_{E,\text{elongated}}(t) = \frac{1}{N_E(t)} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(2\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(2\varphi_i(t)) \right]^2}. \quad (1)$$

The global polar order is defined as

$$P_{E,\text{elongated}}(t) = \frac{1}{N_E(t)} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(\varphi_i(t)) \right]^2}. \quad (2)$$

We also calculate the parameters normalized by the total number of cells:

$$Q_{E,\text{total}}(t) = \frac{1}{N} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(2\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(2\varphi_i(t)) \right]^2}, \quad (3)$$

and

$$P_{E,\text{total}}(t) = \frac{1}{N} \sqrt{\left[\sum_{i \in \mathcal{E}(t)} \sin(\varphi_i(t)) \right]^2 + \left[\sum_{i \in \mathcal{E}(t)} \cos(\varphi_i(t)) \right]^2}. \quad (4)$$

The phenotypic composition is characterized by

$$f_E(t) = \frac{N_E(t)}{N}. \quad (5)$$

These quantities satisfy

$$P_{E,\text{total}}(t) = f_E(t) P_{E,\text{elongated}}(t), \quad Q_{E,\text{total}}(t) = f_E(t) Q_{E,\text{elongated}}(t). \quad (6)$$

When $N_E(t) = 0$, the order parameters normalized by the elongated population are set to zero.

1.1 Time evolution of order parameters

We first analyze the time evolution of the order parameters.

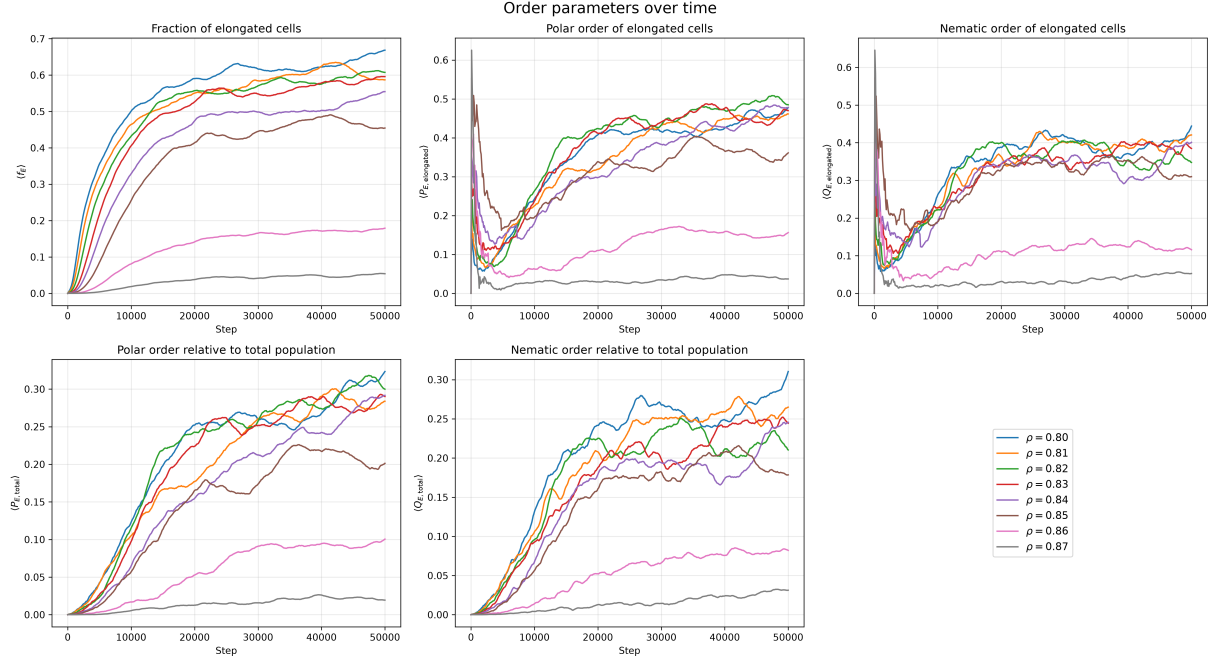


Figure 1: Time evolution of the elongated fraction and orientational order parameters for the different densities. Curves are averaged over independent realizations.

f_E increases progressively from the initially round configuration. For lower densities, a substantial fraction of the population becomes elongated, while the elongated fraction is strongly suppressed at the highest densities.

$P_{E,\text{total}}$ and $Q_{E,\text{total}}$ show the same suppression at high density, indicating that collective orientational order becomes macroscopically small when elongation is inhibited.

$P_{E,\text{elongated}}$ and $Q_{E,\text{elongated}}$ remain finite whenever an elongated subpopulation is present. At very early times, these quantities should be interpreted with caution because N_E is still very small.

1.2 Steady state

To obtain a more stable estimate of the order parameters and compare their dependence on density, we average each observable over the final 10% of each realization. We first perform the temporal average independently for each seed and then average over realizations.

At this stage, the final 10% is used as a late-time averaging window rather than assuming that all realizations have reached a fully stationary state. Longer simulations will be used to assess convergence.

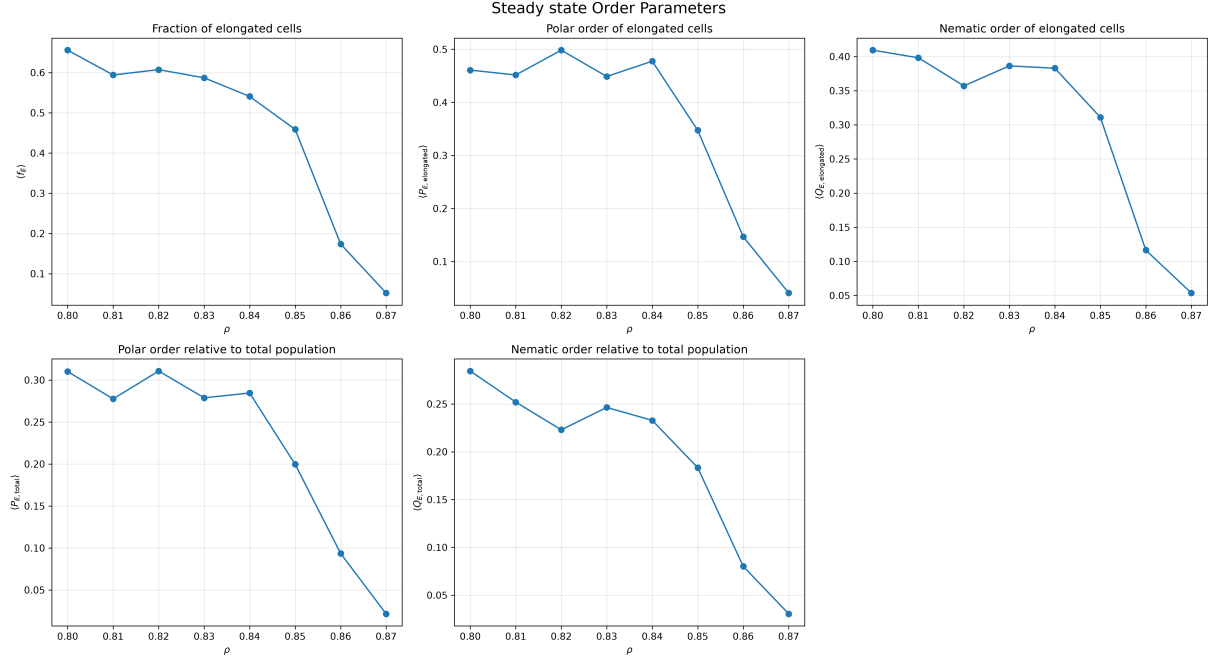


Figure 2: Late-time order parameters as a function of density. Each point is obtained by first averaging over the final 10% of each realization and then averaging over realizations.

The late-time f_E remains relatively large at lower densities and decreases sharply between $\rho = 0.85$ and $\rho = 0.86$. A similar decrease is observed for $P_{E, \text{total}}$ and $Q_{E, \text{total}}$.

Among the quantities considered here, f_E provides the most direct measure of the morphological state of the system and will therefore be used as the primary observable to characterize the transition.

1.3 Susceptibility

To quantify the fluctuations of the elongated fraction, we define its susceptibility as

$$\chi_{f_E} = N_{\text{actual}} \left[\langle f_E^2 \rangle - \langle f_E \rangle^2 \right]. \quad (7)$$

The temporal moments are first evaluated within the late-time window for each realization and are subsequently averaged over seeds.

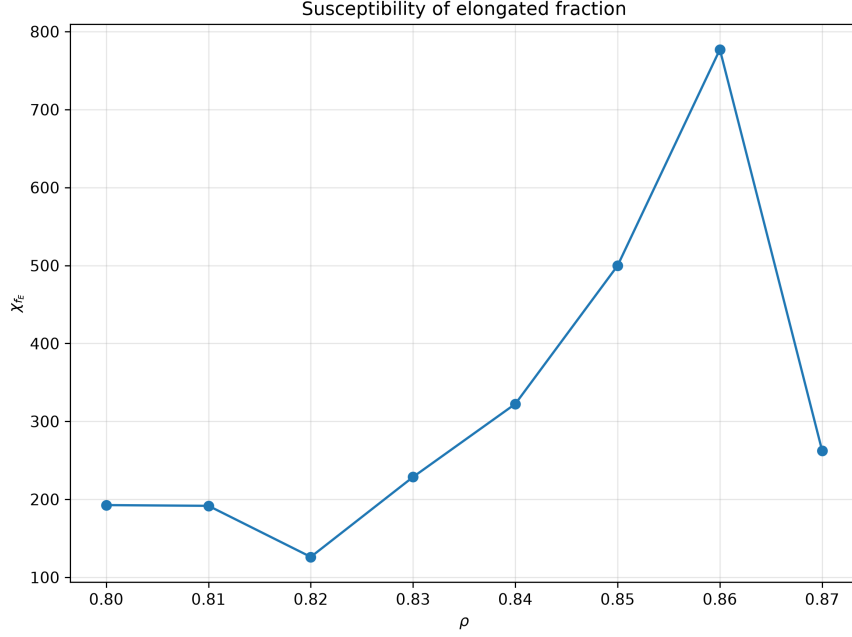


Figure 3: Susceptibility of the elongated fraction as a function of density for $N = 10000$.

For $N = 10000$, the susceptibility displays a pronounced maximum at $\rho = 0.86$, coinciding with the region where the late-time elongated fraction decreases sharply. We therefore use the susceptibility maximum as a reference point to inspect the fluctuations and the distribution of f_E near the transition.

1.4 Elongated-fraction distributions near the susceptibility peak

To investigate the origin of the enhanced fluctuations, we examine the distribution of the late-time averaged elongated fraction, $\langle f_E \rangle_t$, across independent realizations. We consider the density corresponding to the susceptibility maximum together with the immediately neighboring sampled densities.

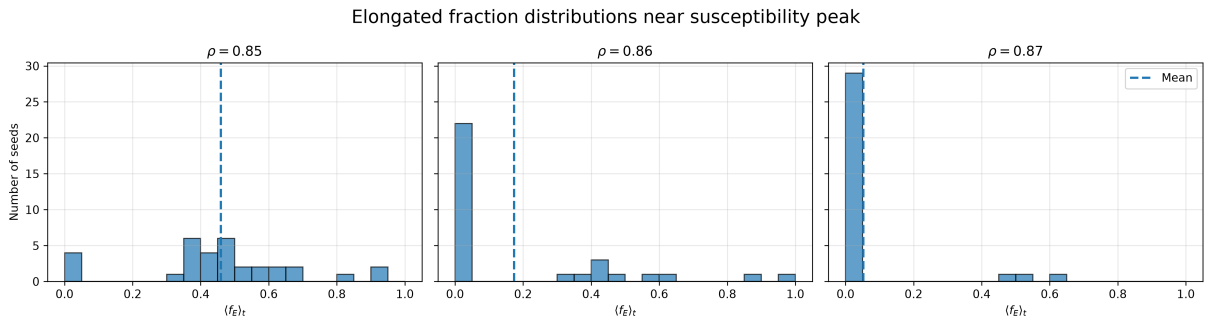


Figure 4: Distribution across independent realizations of the late-time averaged elongated fraction near the susceptibility maximum.

At $\rho = 0.85$, most realizations retain a substantial elongated population, although realizations with very small f_E are already present. At $\rho = 0.86$, realizations with $f_E \approx 0$ coexist with others retaining a finite elongated fraction. At $\rho = 0.87$, most realizations belong to the low- f_E branch, while a small number remain strongly elongated.

1.5 Individual realizations near the susceptibility peak

To determine whether the broad distribution near the susceptibility maximum originates from distinct dynamical behaviors, we inspect the time evolution of f_E for each independent realization at $\rho = 0.86$.

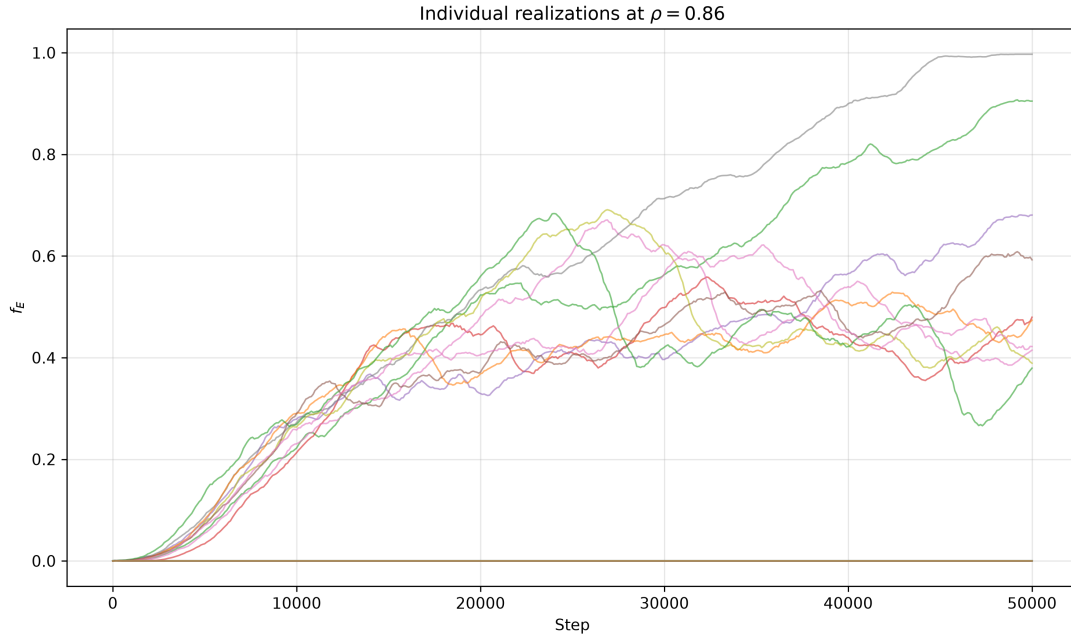


Figure 5: Time evolution of the elongated fraction for individual realizations at $\rho = 0.86$.

The individual trajectories show a strong separation between realizations. Several remain close to the fully round state, whereas others develop and retain a substantial elongated population.